

Machine Learning for Forecasting Hazardous Air Quality Days in Indian Cities

Project Proposal

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1. Introduction and Motivation

Air pollution in Indian cities regularly reaches concentrations classified as hazardous to human health. The public response to such episodes, including health advisories, restrictions on outdoor activity, temporary school closures and distribution of protective equipment, depends on knowing that a severe day is coming before it arrives. A forecast issued on the morning of a hazardous day is of limited value; a forecast issued one to three days ahead can change behaviour.

This project treats that as a supervised machine learning problem. Daily pollution series carry pronounced structure that can be encoded as predictive features: persistence from one day to the next, a weekly cycle driven by traffic and industrial activity, a strong annual cycle driven by meteorology and seasonal agricultural burning, and dependence on recent local conditions. The project constructs such features, trains a range of machine learning models to predict concentrations and threshold exceedances, and evaluates them by the criterion that matters operationally: whether a hazardous day is correctly anticipated.

My interest in this question is also personal. Having grown up in India, air pollution was not an abstract statistic but part of ordinary life: seasonal smog that turned daylight hazy, and days when going outside meant a sore throat. Whether those days can be predicted reliably enough to be worth warning people about is a question I have a genuine stake in.

2. Research Questions

RQ1. How accurately can daily pollution concentrations in a major Indian city be forecast one to three days ahead by machine learning models trained on lagged and calendar features, and how do these compare against classical statistical and naive benchmarks?

RQ2. Is it more effective to forecast the concentration and then apply a health threshold, or to predict the exceedance event directly as a classification problem? These are distinct formulations of the same operational question and need not favour the same model.

RQ3. Treating the forecast as a warning system, what hit rate and false alarm rate does each approach achieve, and how does the choice of decision threshold trade one against the other?

3. Data

Daily mean concentrations of PM_{2.5} and the Air Quality Index for one major Indian city, most likely Delhi, over a period of several years. The Central Pollution Control Board publishes daily AQI bulletins, and the same underlying station data is available through the OpenAQ platform, which provides a free research API. Both sources are public and require no payment or licence. Daily meteorological variables, including temperature, wind speed and humidity, will be introduced as additional predictors where available, since dispersion conditions are a known driver of pollution episodes.

Real monitoring records contain missing observations arising from instrument downtime and reporting gaps. Missingness will be characterised explicitly and handled by a documented strategy, since tree-based learners and neural models differ in their tolerance for absent inputs, and the chosen treatment must not allow information from the future to influence past predictions.

4. Proposed Methodology

Step 1. Exploratory analysis

Characterise the series before modelling it: assess stationarity, examine the autocorrelation structure, and estimate the spectral density to confirm the weekly and annual periodicities quantitatively rather than by assumption. These findings determine which features are worth constructing.

Step 2. Feature construction

Transform the series into a supervised learning problem in which each observation comprises predictors available before the forecast date and a target to be predicted. Candidate features include lagged concentrations at one, two, three and seven days; rolling means and standard deviations over recent windows; calendar indicators for day of week and month; harmonic terms encoding the annual cycle; and meteorological variables where available. Because the informative features are those suggested by the exploratory analysis, this step is a statistical exercise rather than a mechanical one, and its choices will be documented and justified.

Step 3. Model suite

Several model families will be trained on the same feature representation so that comparisons are fair. Benchmarks comprise a naive persistence forecast, a seasonal-naive forecast, and a seasonal ARIMA model, the last retained because a classical statistical benchmark is what gives the machine learning results interpretable meaning. The machine learning suite comprises penalised linear regression; random forests (Breiman, 2001); gradient-boosted trees, using XGBoost (Chen and Guestrin, 2016) or LightGBM; and recurrent neural networks of the LSTM (Hochreiter and Schmidhuber, 1997) or gated recurrent unit type, which operate on the sequence directly rather than on engineered features. Subject to time, a Kolmogorov-Arnold Network will be explored as a recent alternative architecture.

Step 4. Two formulations of the warning problem

The same models will be applied under two formulations. In the regression formulation the predicted concentration is compared against a health threshold to produce a warning. In the

classification formulation the exceedance indicator is predicted directly, which reframes the task as imbalanced binary classification, since hazardous days are by definition a minority class. Class weighting and decision-threshold tuning will be applied, and performance assessed using precision-recall analysis in addition to conventional accuracy measures.

Step 5. Evaluation

All models will be evaluated on held-out data using rolling-origin forecasting at horizons of one to three days, with strict temporal separation between training and evaluation so that no future information can influence a past forecast; standard random cross-validation is inappropriate for dependent data (Bergmeir and Benitez, 2012). Point accuracy will be reported as root mean squared error and mean absolute error. Warning performance will be summarised as a contingency table of hits, misses and false alarms, with hit rate, false alarm rate and a skill score, and the decision threshold will be varied to trace the trade-off between missed hazardous days and unnecessary alarms. Where models permit, feature importance measures will be used to interpret which predictors carry the signal.

5. Expected Contribution

The individual algorithms are established, and the contribution is not the invention of a method. It lies in a controlled comparison across model families for a specific public health decision problem, and in evaluating those models by exceedance-warning performance rather than average error alone. A model with respectable root mean squared error may nonetheless be a poor warning system if it systematically misses extreme days, and separating these two notions of accuracy is the substantive point of the project. Evidence from large-scale forecasting competitions (Makridakis, Spiliotis and Assimakopoulos, 2020) indicates that sophisticated methods do not invariably outperform simpler ones on series of modest length, so a finding in either direction, provided it rests on a rigorous evaluation, is informative.

6. Limitations

- The analysis is confined to a single city, so conclusions may not transfer to cities with different meteorology or emission sources. A multi-city comparison is left as future work.
- The dataset comprises a few thousand daily observations, which is modest for neural models and limits the complexity that can be fitted without overfitting.
- Exceedances are rare by construction, so estimates of warning performance at the most severe thresholds will rest on relatively few events and carry wide uncertainty.
- The models are statistical rather than physical and do not represent atmospheric dispersion, so forecasting skill is expected to decline appreciably at longer horizons.
- Monitoring data is subject to instrument error and gaps, which the chosen missingness treatment can accommodate but cannot correct.

Indicative References

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